

Mahmoud Rabie

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Available for Co-op: September – December 2026 (4 months)

Education

University of British Columbia

Expected April 2029

Bachelor of Applied Science – Electrical and Electronics Engineering

Dean's List

Experience

Undergraduate Research Assistant

May 2026 – Present

UBC Faculty of Applied Science

- Conduct research in quantum networking and next-generation communications protocols, contributing to an industry-partnered national telecom infrastructure initiative
- Mine IEEE INFOCOM and ACM SIGCOMM publications to identify novel research directions; produce technical proposals and literature reviews
- Awarded the **International Undergraduate Student Research Award (IUSRA)** – recognized for research in fiber-optic quantum communications and next-generation telecom protocols; selected among few recipients across UBC's Faculty of Engineering, valued at \$10,000

Al-Nisf Electrical Company | AutoCAD, MS Office

May 2025 – July 2025

Electrical and Power Engineering Intern

- Supported the design of commercial power distribution systems using **AutoCAD**, gaining exposure to real-world electrical layouts and standards
- Verified on-site electrical installations by inspecting wiring and switchboards for compliance with engineering and safety requirements

Projects

Recycla – AI Recycling Platform | Python, TensorFlow/Keras, React, Raspberry Pi

2025 – 2026

- Built an end-to-end AI waste classification system: trained a **MobileNetV2** transfer-learning model across 5 material classes (~5,000 images) using **TensorFlow/Keras** in Google Colab, achieving high accuracy under varied lighting
- Integrated the model with servo-actuated **Raspberry Pi** sorting hardware and deployed a public-facing web platform with real-time classification feedback and a full project website

UBC Okanagan Steel Bridge Team Website | HTML, CSS3, Git

January 2026 – Present

- Built a responsive, single-page website using semantic **HTML5** (header/nav/main/section/footer) and reusable page sections to improve maintainability
- Structured "sponsor" and "team" components with consistent HTML patterns (card layout, predictable class naming) to enable fast content updates without breaking layout; debugged HTML nesting issues affecting cross-browser layout using **DevTools**

UBC Workday Schedule Tool | JavaScript, Chrome Extension APIs (MV3), OAuth2

2025

- Engineered the Google Calendar integration end-to-end: implemented **OAuth2** authorization flow and **Google Calendar API** event creation to export UBC course schedules from Workday directly to users' Google Calendars
- Developed backend processing logic for parsing and transforming Workday course data; architected **chrome.storage** state management and contributed to the Manifest V3 service worker and content script pipeline

Activities

Aerospace and Rocketry Club | C++, Arduino IDE, Microcontrollers

November 2025 – Present

Avionics Subteam Member

- Prototyping an embedded system for a rocket using an **ESP32 microcontroller**, sensors, and **LoRa** telemetry to support ground and flight data transmission
- Testing and refining sensor signal paths and power routing on breadboard implementations to inform custom **PCB design**
- Contributing to PCB planning and schematic development aimed at minimizing wiring, improving reliability, and enabling modular system integration

Technical Skills

Languages: MATLAB, C, C++, Python, HTML, SQL, JavaScript, TypeScript

Skills & Tools: Linux, Bash, Git, SolidWorks, AutoCAD, MS Office, Microcontrollers, Verilog HDL, Wiring, TensorFlow, React, Raspberry Pi, Docker

Relevant Coursework: Numerical Methods of Analysis, Logic Design, Signals and Communication Systems, Computer Architecture and Design